Statistical Modelling in Public Health

What It Is · How to Build It · How to Report It

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"All models are wrong, but some are useful." — George Box

Course Outline

01 What is Statistical Modelling?

02 Types of Models in Public Health

03 Steps in Model Building

04 Assumptions & Diagnostics

05 Interpretation & Reporting

SECTION 1

What is Statistical Modelling?

Turning data into decisions that protect population health

What is Statistical Modelling?

A statistical model is a mathematical representation of the relationship between variables, used to describe, explain, predict, or control outcomes in a population.

DESCRIBE

Summarise how variables relate
(e.g. mean BP by age group)

EXPLAIN

Quantify the effect of an exposure
after adjusting for confounders

PREDICT

Estimate risk scores or disease
probability for new individuals

CONTROL

Identify targets for intervention
and evaluate programme impact

Why Do We Model in Public Health?

Exposure–outcome complexity

Health outcomes are shaped by dozens of social, biological, and environmental factors acting simultaneously. Models let us isolate and quantify individual effects.

Confounding is ubiquitous

Observational data — the norm in public health — is riddled with confounders. Regression adjusts for these so we can approach causal inference.

Small effects, large samples

Many public health exposures have modest effect sizes. Multivariable models increase statistical power by reducing unexplained variance.

Policy & resource allocation

Risk prediction models (e.g. cardiovascular risk scores) directly inform clinical guidelines, screening criteria, and health resource planning.

Programme evaluation

Interrupted time-series, difference-in-differences, and propensity score models let us evaluate interventions without RCT infrastructure.

What Is a Statistical Model? (Ch. 18)

A statistical model is a mathematical expression of the assumed relationship between one or more **explanatory variables** and a **response variable**. *-- Kirkwood & Sterne, p. 171*

SYSTEMATIC COMPONENT

The deterministic part: the average (expected) value of Y given the values of predictor variables $X_1, X_2 \ldots X_k$.

Expressed as: $E(Y|X) = \beta_0 + \beta_1 X_1 + \ldots + \beta_k X_k$

RANDOM COMPONENT

The stochastic part: residual variation around the expected value, assumed to follow a probability distribution (Normal, Binomial, Poisson, etc.).

LINK FUNCTION

In Generalised Linear Models (GLMs), a function $g(\cdot)$ that links $E(Y)$ to the linear predictor η.

Identity (linear), logit (logistic), log (Poisson).

The Generalised Linear Model (GLM) Framework (Ch. 18)

$$g(E[Y]) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k$$

Regression Model	Distribution	Link Function	Parameter	Outcome Scale
Linear Regression	Normal	Identity: $g(\mu)=\mu$	Mean (μ)	Continuous
Logistic Regression	Binomial	Logit: $g(\pi)=\log(\pi/1-\pi)$	Probability (π)	Binary 0/1
Poisson Regression	Poisson	Log: $g(\lambda)=\log(\lambda)$	Rate (λ)	Count/Rate
Cox (Survival)	None	Log-log (partial likelihood)	Hazard h(t)	Time-to-event

Kirkwood & Sterne – Medical Statistics (2nd ed.) | Chapter 18, Table 18.1 (pp. 172–173)

SECTION 2

Types of Models in Public Health

Matching the model to the outcome type and research question

Variable Types & Their Roles in Models (Ch. 18–19)

RESPONSE / OUTCOME (Y)

Continuous

Blood pressure, height, weight → Linear regression

Binary

Disease yes/no, died yes/no → Logistic regression

Count/Rate

Number of events, incidence rate → Poisson regression

Time-to-event

Survival time, time to relapse → Cox regression

EXPLANATORY VARIABLES (X)

Continuous predictor

Age, BMI, cholesterol — enter as numeric

Binary/categorical

Sex, treatment group — enter as dummy (0/1) variables

Ordinal

Disease severity — can use as numeric or dummy coded

Interaction terms

$X_1 \times X_2$ — tests whether effect of X_1 varies by X_2

Choosing the Right Model

Regression Model	Outcome Type	Link fn	PH Use	Example Outcome
Linear Regression	Continuous	Identity	GLM	Systolic BP, BMI, cholesterol levels
Logistic Regression	Binary (yes/no)	Logit	GLM	Disease presence, mortality, smoking status
Poisson / Negative Binomial	Count or Rate	Log	GLM	Hospital admissions, disease incidence rates
Cox Proportional Hazards	Time-to-event	Log-log	GLM	Survival time, time to diagnosis/relapse
Multinomial Logistic	Nominal (>2 levels)	Logit	GLM	Disease stage, treatment group allocated
Ordinal Logistic	Ordinal (>2 levels)	Logit	GLM	Likert scales, pain score, cancer grading

SECTION 3

Steps in Model Building

A systematic workflow from question to conclusion

Model Building Principles

1

Define Research Question

Clearly specify outcome (Y) and main exposure (X). State whether goal is prediction or causal inference.

2

Select Candidate Variables

Use subject-matter knowledge, directed acyclic graphs (DAGs), and prior literature. Avoid data-driven selection only.

3

Check Assumptions

Linearity, independence, homoscedasticity, normality of residuals (OLS). Check multicollinearity (VIF < 10).

4

Build & Assess Model

Crude model → adjusted model. Compare with likelihood ratio test, AIC/BIC, R^2, or pseudo-R^2.

5

Interpret & Report

Report coefficients, ORs, or HRs with 95% CI. Describe confounding adjustment and model fit.

Model Building Strategy (Ch. 18–19)

The choice of variables to include in a model should be guided primarily by **subject-matter knowledge** *rather than by statistical criteria alone. -- Kirkwood & Sterne, p. 180*

1

Define question & outcome

Specify Y, the main exposure X, and the study design (Ch. 18). Draw a DAG to map causal relationships.

2

Select candidate variables

Use prior literature + clinical knowledge. Apply the events-per-variable (EPV) rule: ≥10 events per covariate.

3

Explore data (EDA)

Distributions, missingness, outliers, linearity of continuous predictors (Ch. 19). Tabulate outcome by exposure.

4

Fit unadjusted (crude) model

Model Y on X alone. Obtain crude estimate (β, OR, HR). This is the starting reference for confounding assessment.

5

Fit adjusted (multivariable) model

Add covariates. Check 10% change rule: if coefficient changes ≥10% on adding a variable → confounder → retain.

6

Check assumptions & model fit

Residual plots, goodness-of-fit tests, AIC/BIC, likelihood ratio tests. See Chapters 20–25 for model-specific checks.

The Model Building Workflow — Overview

1. Define the question
2. Select variables
3. Explore the data
4. Choose model type
5. Fit crude model
6. Fit adjusted model
7. Check assumptions
8. Evaluate model fit
9. Interpret results
10. Report & communicate

Steps 1–3: Defining the Problem & Exploring Data

STEP 1 — Define the Research Question

- State the outcome variable Y (what are you modelling?)
- Identify the main exposure/predictor of interest X
- Clarify the goal: description, explanation, or prediction?
- Consider study design: cross-sectional, cohort, case-control
- Draw a Directed Acyclic Graph (DAG) to map hypothesised relationships

STEP 2 — Variable Selection

- Use prior literature + clinical/epidemiological knowledge
- Select potential confounders from the DAG — not from p-values
- Include effect modifiers (interaction terms) if biologically plausible
- Avoid: stepwise selection, $p < 0.05$ as sole criterion
- Rule of thumb: ≥ 10 events per variable (EPV rule)

STEP 3 — Exploratory Data Analysis (EDA)

- Check distributions: histograms, box plots, Q-Q plots
- Examine missingness: MCAR / MAR / MNAR — consider multiple imputation
- Assess linearity of continuous predictors (fractional polynomials / restricted cubic splines)
- Compute correlation matrix — flag multicollinearity ($r > 0.8$)
- Tabulate outcome by exposure (Table 1)

Steps 4–6: Choosing & Fitting the Model

4

Choose Model Type — Match the model to the outcome distribution

Continuous Y → Linear | Binary Y → Logistic | Count Y → Poisson | Time-to-event Y → Cox | Correlated data → Mixed/GEE

STEP 5 — Fit Crude (Unadjusted) Model

- Regress Y on the main exposure X only
- Provides the unadjusted effect estimate (crude OR/β/HR)
- Always report alongside adjusted model
- Basis for 10% confounding rule (see below)

Crude β_1: Effect of X on Y ignoring all confounders

STEP 6 — Fit Adjusted (Multivariable) Model

- Add confounders and covariates identified in Step 2
- Adjusted β_1 represents the independent effect of X
- If $|\%\Delta\beta_1| \geq 10\%$ → confounding confirmed; retain variable
- Consider biological plausibility — keep confounders even if $p > 0.05$

Adjusted β_1: Effect of X on Y holding all other variables constant

Modelling a continuous outcome — Chapters 20 & 21

Simple Linear Regression (Ch. 20)

$$Y = \beta_0 + \beta_1 X + \varepsilon \quad \text{where } \varepsilon \sim N(0, \sigma^2)$$

Parameter Interpretation

β_0 (Intercept)
Predicted value of Y when X = 0. May not always have clinical meaning.

β_1 (Slope)
Average change in Y for each 1-unit increase in X, holding all other variables constant.

σ^2 (Residual variance)
Spread of observations around the regression line. Estimated by the mean squared error (MSE).

R^2 (Coefficient of determination)
Proportion of variation in Y explained by the model. Ranges 0–1; higher = better fit.

K&S Example: Birthweight & Gestational Age

Research question: Does gestational age (X, weeks) predict birthweight (Y, kg)?

Model: Birthweight = $\beta_0 + \beta_1$(Gestational age) + ε

Fitted: $\hat{Y} = -5.26 + 0.197 \times$ (Gestational age)

Interpretation: For each additional week of gestation, birthweight increases by 0.197 kg on average.

(Kirkwood & Sterne, Ch. 20, pp. 196–202)

Multiple Linear Regression (Ch. 20–21)

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k + \varepsilon$$

Adjusted coefficients

Each β represents the effect of X_j on Y after adjusting for all other predictors in the model. This is the key advantage over simple regression.

Dummy variables

Categorical predictors with k levels are represented by k−1 dummy (0/1) variables. The omitted category is the reference group. (K&S pp. 204–207)

Interaction terms

Add a product term $X_j \times X_m$ to test whether the effect of X_j on Y differs across levels of X_m. Significant interaction = effect modification. (Ch. 21)

Partial F-test / ANOVA

Used to compare nested models. Tests whether a group of predictors together significantly improve model fit (K&S pp. 214–216).

Multicollinearity

High correlation between predictors inflates standard errors. Diagnose with VIF > 10. Consider removing redundant variables or using ridge regression.

R^2 vs Adjusted R^2

Adjusted R^2 penalises for the number of predictors, making it suitable for comparing models with different numbers of variables (K&S p. 213).

Assumptions & Diagnostics: Linear Regression (Ch. 20)

Assumption	How to Check	If Violated
Linearity	Residuals vs. fitted plot — no systematic curve	Transform X (log, square root) or use polynomial terms
Independence	Inspect study design — independent subjects?	Clustered data → multilevel models or GEE (Ch. 29)
Homoscedasticity	Scale-location plot — constant spread around zero	Weighted least squares; transform Y (log Y)
Normality of residuals	Normal Q-Q plot — points on the diagonal line	Large samples: CLT protects. Small n: transform or non-parametric
No multicollinearity	VIF < 10; pairwise correlations < 0.8	Remove one of two correlated variables; principal components
No influential outliers	Cook's distance < 4/n; leverage vs. residuals plot	Investigate observation — data error or genuine outlier?

Reporting Linear Regression Results (Ch. 20–21)

Example: Predictors of systolic blood pressure (SBP) — Multiple linear regression

Variable	β (Crude)	95% CI	β (Adjusted)	95% CI	p-value
Age (per 10 yrs)	+3.8	2.5 to 5.1	**+3.2**	2.0 to 4.4	< 0.001
Sex: Male (ref: Female)	+6.2	3.8 to 8.6	**+5.1**	2.7 to 7.5	< 0.001
BMI (per 1 kg/m²)	+1.1	0.7 to 1.5	**+0.9**	0.5 to 1.3	< 0.001
Smoking (current vs never)	+4.3	1.5 to 7.1	+3.1	0.4 to 5.8	0.024
Physical activity (active)	−3.6	−5.8 to −1.4	−2.8	−5.0 to −0.6	0.013

Model R^2 = 0.34 (34% of variance in SBP explained). All β are in mmHg. Adjusted for all variables shown. n = 820.

Report: n, R^2, adjusted R^2, regression coefficients with 95% CIs, p-values, which variables were adjusted for, and how assumptions were checked (STROBE checklist)

Binary outcomes and odds ratios — Chapters 22 & 23

Logistic Regression: Rationale & Model (Ch. 22)

Why NOT Linear Regression?

When Y is binary (0/1):
- Predicted values can fall outside [0, 1]
- Residuals are not normally distributed
- Variance is not constant (heteroscedastic)

→ We model the **log-odds (logit)** of the outcome instead of Y itself.

The Logistic Model

$$\log[\pi/(1-\pi)] = \beta_0 + \beta_1 X_1 + \ldots + \beta_k X_k$$

where $\pi = P(Y = 1 \mid X)$

Back-transform: $\pi = e\eta / (1 + e\eta)$
where $\eta = \beta_0 + \beta_1 X_1 + \ldots + \beta_k X_k$

The Odds Ratio (OR) — Key Output of Logistic Regression

OR = e^β_1 → For a 1-unit increase in X_1, the **odds** of the outcome are multiplied by e^β_1

OR = 1.0: No association | **OR > 1:** Increased odds | **OR < 1:** Decreased odds

K&S Example (Ch. 22): OR = 2.4 for smoking and lung disease → Smokers have 2.4 times the odds of lung disease compared to non-smokers, adjusting for age and sex.

Multiple Logistic Regression (Ch. 22–23)

Adjusted Odds Ratios

Each OR is adjusted for all other variables in the model.
Interpretation: the OR for X_1 is the effect of X_1 on the odds of Y, holding X_2, X_3, … X_k constant.
This controls for confounding (K&S p. 236).

Handling Categorical Predictors

Use k−1 dummy variables for a factor with k categories.
Choose a reference category (usually the lowest risk or most common group).
Each OR compares that category to the reference.

OR vs. Relative Risk (RR)

OR ≈ RR when outcome is rare (< 10%).
When outcome is common, OR overestimates RR.
For common outcomes: use log-binomial regression to estimate RR directly (K&S pp. 246–247).

Maximum Likelihood Estimation

Logistic regression parameters are estimated by maximising the likelihood function — the probability of observing the data given the model.
No closed-form solution: uses iterative numerical methods (Newton-Raphson).

Assessing Logistic Regression Model Fit (Ch. 22–23)

Likelihood Ratio Test (LRT)

Compares the log-likelihood of the full model vs. a nested reduced model.
2(log L_full − log L_reduced) ~ χ^2 with df = number of extra parameters.
Significant LRT → added variables significantly improve fit (K&S p. 240).

Hosmer–Lemeshow Test

Divides data into ~10 groups by predicted probability. Compares observed vs. expected number of events in each group.
$p > 0.05$ = acceptable calibration. Can be insensitive with small samples (K&S p. 242).

AUC / C-statistic

Area under the ROC curve — measures the model's discrimination ability (ability to rank a case higher than a control).
0.5 = no discrimination; 0.7–0.8 = acceptable; > 0.8 = excellent (K&S p. 243).

AIC / BIC

Akaike (AIC) and Bayesian (BIC) Information Criteria penalise model complexity.
Lower values indicate a better-fitting, more parsimonious model.
Useful for comparing non-nested models.

Pseudo-R^2 (Nagelkerke)

Approximates the proportion of variation explained. Not directly comparable to OLS R^2.
Use only for comparing models on the same dataset. Report alongside LRT (K&S p. 241).

Residuals & Influence

Pearson residuals, deviance residuals, and influence statistics (DFBETA, leverage).
Plot against covariate values to detect outliers and influential observations (K&S pp. 244–245).

Presenting Logistic Regression Results (Ch. 22)

Example: Risk factors for low birthweight (<2.5 kg) — K&S Chapter 22 dataset

Variable	n (%)	Crude OR (95% CI)	p	Adjusted OR (95% CI)	p
Maternal age < 20 yrs	87 (18%)	2.1 (1.3–3.4)	0.003	1.8 (1.1–3.0)	0.020
Smoking in pregnancy	143 (29%)	2.8 (1.9–4.1)	<0.001	2.4 (1.6–3.6)	<0.001
No antenatal care	98 (20%)	3.5 (2.2–5.5)	<0.001	2.9 (1.8–4.7)	<0.001
Low maternal weight (<50kg)	112 (23%)	2.0 (1.3–3.1)	0.002	1.7 (1.1–2.7)	0.024
Male infant (ref: Female)	248 (51%)	1.2 (0.8–1.7)	0.36	1.1 (0.8–1.6)	0.52

n = 490; 128 cases of low birthweight (26%). Model: Hosmer–Lemeshow p = 0.42; AUC = 0.76; Nagelkerke R^2 = 0.22. Adjusted for all variables listed.

K&S Note: Always check whether OR approximates RR — here outcome prevalence = 26%, so OR overestimates RR. Consider log-binomial model.

Incidence rates, count outcomes, and rate ratios — Chapter 24

Poisson Regression: Rates & Counts (Ch. 24)

$\log(\lambda) = \beta_0 + \beta_1 X_1 + \ldots + \beta_k X_k$ | Rate Ratio (RR) = e^{β_j}

When to Use Poisson Regression

Use when the outcome is:
- A count of events (number of hospital admissions, seizures, injuries)
- An incidence rate (events per person-year)
- A proportion with a common outcome (as alternative to logistic regression)

Key assumption:
- Mean = Variance (equidispersion). If variance > mean → overdispersion → use Negative Binomial or robust SEs

Offset term:
- Include log(person-time) as an offset to model rates:
  $\log(\lambda) = \log(PT) + \beta_0 + \beta_1 X_1 + \ldots$ (K&S p. 268)

K&S Example: Child Mortality Rates

Question: Does maternal education predict under-5 mortality rate?

Outcome: Number of child deaths per 1000 child-years

Model:
log(deaths) = log(child-years) + β_0 + β_1(mother educ.) + β_2(wealth)

Result: RR = 0.67 (95% CI: 0.52–0.86) for secondary vs. no education
→ Children of mothers with secondary education have 33% lower mortality rate.

(K&S Ch. 24, pp. 265–278)

Overdispersion & Negative Binomial Regression (Ch. 24)

Overdispersion occurs when the observed variance in count data exceeds the mean — violating Poisson's core assumption. Causes: excess zeros, clustering, unobserved heterogeneity. (K&S p. 274)

Detecting Overdispersion

1. Pearson χ^2 / df >> 1 (> 1.5 suggests overdispersion)
2. Deviance / df ratio > 1.5
3. Negative Binomial fits significantly better than Poisson (LRT)
4. Compare AIC: Neg. Binomial should show lower AIC

Solutions: Negative Binomial

Adds a dispersion parameter α to account for extra-Poisson variation.
$Var(Y) = \mu + \alpha \cdot \mu^2$ ($\alpha = 0 \rightarrow$ reduces to Poisson)
All interpretations remain as rate ratios (e^{β}).
Most appropriate when overdispersion is substantial.

Solutions: Robust Standard Errors

Apply sandwich (Huber-White) variance estimator.
Coefficients unchanged; SEs are corrected for overdispersion.
Quicker fix; less formally tests for overdispersion.
K&S recommend examining both approaches (p. 276).

Zero-Inflated Models

When excess zeros exist (two data-generating processes):
- ZIP: Zero-Inflated Poisson
- ZINB: Zero-Inflated Negative Binomial

Model the probability of structural zeros separately from the count process.

Using Poisson to Estimate Relative Risks (Ch. 24)

When the binary outcome is **common (> 10%)**, logistic regression overestimates the relative risk. Poisson regression with robust standard errors estimates the **Relative Risk (RR)** *directly — preferred in cross-sectional studies. (K&S pp. 246, 278)*

	LOGISTIC (gives OR)	POISSON + robust SE (gives RR)	LOG-BINOMIAL (gives RR)
Model:	$\text{logit}(\pi) = \beta_0 + \beta_1 X$	$\log(\pi) = \beta_0 + \beta_1 X$	$\log(\pi) = \beta_0 + \beta_1 X$ (Binomial family)
Measure:	Odds Ratio (OR = e^β)	Relative Risk (RR = e^β)	Relative Risk (RR = e^β)
When:	Rare outcomes (< 10%) Case-control studies	Common outcomes (> 10%) Cross-sectional studies	Common binary outcomes
Note:	OR overestimates RR when outcome is common	Preferred: directly interpretable as risk ratio	Can fail to converge; Poisson + robust SE is more stable

Time-to-event outcomes and hazard ratios — Chapter 25

Survival Analysis: Key Concepts (Ch. 25)

Survival Time (T)

The time from a defined origin (entry, diagnosis, randomisation) to the event of interest or censoring.
Must be unambiguously defined in the study protocol.

Censoring

Observation ends before the event occurs:
• Right-censored: lost to follow-up, withdrawn, or study ends
Assumption: censoring is non-informative (independent of event) — K&S p. 293.

Survival Function S(t)

$S(t) = P(T > t)$ — the probability of surviving beyond time t.
Kaplan-Meier estimator (non-parametric) is the standard method.
Log-rank test compares survival between groups.

Hazard Function h(t)

The instantaneous rate of the event at time t, given survival to t.
$h(t) = \lim_{\Delta t \to 0} P(t \leq T < t+\Delta t \mid T \geq t) / \Delta t$
This is what Cox regression models (K&S p. 298).

Cumulative Hazard H(t)

$H(t) = -\log[S(t)]$
Used in model diagnostics: log(H(t)) vs. log(t) should be parallel lines if proportional hazards assumption holds (K&S p. 305).

Person-Time at Risk

Each participant contributes person-time from entry until event or censoring.
Total person-time = sum of all individual follow-up durations.
Used in the denominator of incidence rates.

Cox Proportional Hazards Regression (Ch. 25)

$$h(t|X) = h_0(t) \cdot \exp(\beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k) \quad | \quad HR = e^{\beta_j}$$

Model Features

Semi-parametric:
$h_0(t)$ is an unspecified baseline hazard — no parametric distribution for T assumed. This is the strength of Cox regression.

Partial likelihood:
Parameters estimated using only the rank-ordering of event times, not the actual times. Censored observations contribute information (K&S p. 303).

Hazard Ratio (HR):
$HR = e^{\beta_j}$. Interpreted as the ratio of hazard rates between two groups at any time point, assuming proportional hazards.

HR = 1: No effect | HR > 1: Increased hazard | HR < 1: Protective

PH Assumption & Diagnostics

Proportional Hazards (PH) assumption:
The HR is constant over time — the hazard ratio between groups does not change.

How to check:
- Log-log survival plot: $\log(-\log(S(t)))$ vs. $\log(t)$ — lines should be parallel
- Schoenfeld residuals vs. time — should show no trend
- Include time-dependent covariate ($X \times$ time) — should not be significant

If violated:
- Stratify Cox model by the violating variable
- Use extended Cox model with time-varying covariates
- Consider accelerated failure time (AFT) models (K&S p. 310)

Reporting Cox Regression Results (Ch. 25)

Example: Predictors of time to death in HIV-positive adults on ART — K&S style

Variable	Events/N	Crude HR (95% CI)	p	Adjusted HR (95% CI)	p
CD4 < 200 cells/µL	68/180	3.2 (2.1–4.9)	<0.001	2.7 (1.7–4.2)	<0.001
WHO Stage III/IV	55/140	2.8 (1.8–4.3)	<0.001	2.2 (1.4–3.5)	0.001
Age > 40 yrs	42/115	1.9 (1.2–3.0)	0.005	1.6 (1.0–2.5)	0.042
Male sex (ref: Female)	51/170	1.3 (0.9–1.9)	0.18	1.2 (0.8–1.8)	0.36
On cotrimoxazole prophylaxis	28/180	0.6 (0.4–0.9)	0.018	0.7 (0.4–1.0)	0.056

n = 490; 182 deaths; median follow-up 24 months. PH assumption verified by Schoenfeld residuals (all p > 0.10). Adjusted for all variables listed.

Report: HR with 95% CI, PH assumption check, number of events/total, censoring method, and which covariates were adjusted for (STROBE item 16b)

SECTION 4

Assumptions & Diagnostics

A model that violates its assumptions produces misleading inferences

Assumptions: Linear Regression (OLS)

Assumption	What It Means	How to Check
Linearity	Relationship between X and Y is linear	Residual vs. fitted plot — no pattern
Independence	Observations are independent of each other	Study design — cluster/repeated use MLM
Homoscedasticity	Constant variance of residuals across fitted values	Scale-location plot — flat smooth line
Normality	Residuals are approximately normally distributed	Q-Q plot — points lie on diagonal
No multicollinearity	Predictors are not highly correlated with each other	VIF < 10 (strict: < 5); condition index
No influential outliers	No single observation unduly drives the model	Cook's distance < 4/n; leverage plots

Assumptions & Fit: Logistic Regression

Key Assumptions

Binary outcome

Outcome coded 0/1; no ordered structure assumed

Independence

Observations independent (use GEE for clusters)

Linearity in log-odds

Log-odds of outcome is linear in continuous predictors

No perfect separation

Predictors must not perfectly predict Y (leads to infinite β)

No multicollinearity

VIF < 10; check correlation matrix

Adequate sample

Min 10 events per variable (EPV ≥ 10)

Model Fit Statistics

Hosmer-Lemeshow

Tests calibration: observed vs. expected frequencies across deciles. $p > 0.05$ = good fit

AUC / C-statistic

Discrimination: area under ROC curve. 0.7-0.8 = acceptable; > 0.8 = excellent

Pseudo-R^2 (Nagelkerke)

Approximates proportion of variance explained. Compare between models (not absolute)

AIC / BIC

Information criteria for model comparison. Lower = better fit (penalised for complexity)

Likelihood Ratio Test

Compares nested models. Significant LRT = additional variables improve fit significantly

Systematic Model Checking (K&S Ch. 18–25)

SPECIFICATION

- ✓ Is the outcome distribution correctly specified?
- ✓ Are continuous predictors linear on the link scale?
- ✓ Are all important confounders included? (10% rule)
- ✓ Are interaction terms needed? Test product terms

RESIDUALS

- ✓ Plot residuals vs. fitted values — no pattern
- ✓ Normal Q-Q plot of residuals (linear only)
- ✓ Deviance / Pearson residuals (GLMs)
- ✓ Identify influential observations (Cook's D, DFBETA)

MODEL FIT

- ✓ Likelihood Ratio Test — nested model comparison
- ✓ AIC / BIC — compare competing models
- ✓ Goodness-of-fit: HL test (logistic), deviance (Poisson)
- ✓ ROC/AUC for discrimination (prediction models)

ASSUMPTIONS

- ✓ Proportional hazards — Schoenfeld residuals (Cox)
- ✓ Independence — study design review
- ✓ Overdispersion — Pearson χ^2/df (Poisson)
- ✓ VIF < 10 — multicollinearity (all models)

Common Modelling Pitfalls

⚠ Multicollinearity

Highly correlated predictors inflate standard errors, making coefficients unstable and hard to interpret.

Detect: VIF > 10 (threshold) | correlation matrix
Fix: Remove redundant variable, use PCA, or ridge regression

Missing Data

Complete case analysis (listwise deletion) produces biased estimates if data is not MCAR.

MAR: Multiple imputation (MI) by chained equations
MNAR: Sensitivity analysis; pattern-mixture models
Report: Amount, pattern, and method used

Overfitting

Model fits training data too closely — poor generalisability to new data.

Prevention: Follow EPV ≥ 10 rule; avoid stepwise selection
Validation: Internal (bootstrap, cross-validation) or external cohort

↔ Non-linearity

Treating continuous predictors as linear when the relationship is curved.

Detect: Martingale residuals (Cox); LOWESS smoother
Fix: Restricted cubic splines (RCS) or fractional polynomials

SECTION 5

Interpretation & Reporting

Translating model output into evidence that informs public health decisions

Interpreting Model Coefficients

Linear Regression *β (unstandardised)*	**For every 1-unit increase in X, Y increases/decreases by β units, holding all other variables constant.** *Example: β = 2.3 mmHg → Each 1-year increase in age is associated with a 2.3 mmHg higher systolic BP.* Note: Standardised β compares effect sizes across predictors with different units
Logistic Regression *Odds Ratio (OR = e^β)*	**For every 1-unit increase in X, the odds of the outcome are multiplied by OR, holding all others constant.** *Example: OR = 1.45 → Each 1-unit increase in X is associated with 45% higher odds of disease.* Note: OR ≠ RR (relative risk) when outcome is common (>10%); use log-binomial for RR
Cox Regression *Hazard Ratio (HR = e^β)*	**For every 1-unit increase in X, the instantaneous hazard of the event is multiplied by HR at any given time.** *Example: HR = 0.72 → Treatment reduces the hazard of death by 28% at any point in follow-up.* Note: Assumes proportional hazards — verify with Schoenfeld residuals or log-log plot

Reporting Standards for Multivariable Studies

STROBE (Observational Studies)

- ✓ Study design and setting clearly described
- ✓ Eligibility criteria (inclusion/exclusion)
- ✓ Outcome, exposure, and covariate definitions
- ✓ How confounders were selected and handled
- ✓ Methods for addressing missing data
- ✓ Number of participants at each stage
- ✓ Crude and adjusted estimates with 95% CI
- ✓ Sensitivity analyses and subgroup results

TRIPOD (Prediction Models)

- ✓ Type of model: development or validation
- ✓ Source of data and study period
- ✓ Outcome(s) and predictor(s) defined
- ✓ Sample size and events-per-variable
- ✓ Missing data handling method stated
- ✓ Model development: selection, shrinkage
- ✓ Model performance: calibration + discrimination
- ✓ Limitations: overfitting, generalisability

Presenting Results: The Regression Table

Example: Predictors of hypertension (logistic regression)

Variable	n (%)	Crude OR (95% CI)	p-value	Adjusted OR (95% CI)	p-value
Age (per 10 yrs)	—	1.52 (1.31–1.76)	< 0.001	**1.44 (1.22–1.69)**	< 0.001
Sex: Male (ref: Female)	312 (48%)	1.85 (1.42–2.41)	< 0.001	**1.61 (1.21–2.14)**	0.001
Obese (BMI ≥ 30)	198 (31%)	2.34 (1.78–3.08)	< 0.001	**2.07 (1.55–2.76)**	< 0.001
Current smoker	143 (22%)	1.28 (0.95–1.72)	0.10	**1.18 (0.86–1.62)**	0.31
Physical inactivity	221 (34%)	1.76 (1.35–2.29)	< 0.001	**1.53 (1.16–2.01)**	0.003
Diabetes mellitus	97 (15%)	2.92 (2.08–4.09)	< 0.001	**2.44 (1.71–3.48)**	< 0.001

Adjusted for all variables listed. Bold = statistically significant ($p < 0.05$). n = 645 participants; 218 events.

Reporting Standards: STROBE & K&S Guidance (Ch. 18–25)

STROBE (STrengthening the Reporting of OBservational studies in Epidemiology) provides a **22-item checklist** for transparent reporting of regression analyses in cohort, case-control, and cross-sectional studies.

Title/Abstract	State study design; abstract should give informative summary of results
Study Design	State study design; eligibility criteria; sources and methods for selection
Statistical Methods	Describe all statistical methods; how confounders handled; subgroup analyses; how missing data addressed
Results — Outcome	Report unadjusted AND adjusted estimates; 95% CIs; which covariates adjusted for
Introduction	Scientific background; specific research objectives
Variables	Define outcome, exposures, covariates; sources; measurement methods
Results — Participants	Numbers in each stage; reasons for non-participation; Table 1 characteristics
Discussion	Key results; limitations including bias/imprecision; generalisability

Kirkwood & Sterne – Medical Statistics (2nd ed.) | STROBE Statement (Lancet 2007); K&S Ch. 18–25 reporting guidance

1 — Theory first

Use subject-matter knowledge and DAGs — not stepwise p-values — to select variables (K&S p. 180)

2 — Match model to outcome

Outcome distribution determines the model family and link function (K&S Ch. 18)

3 — Always report crude + adjusted

Show the unadjusted effect alongside the adjusted — transparency on confounding control

4 — Apply the 10% rule

A variable is a confounder if its inclusion changes the main coefficient by ≥ 10% (K&S p. 237)

5 — Check every assumption

Violated assumptions produce biased estimates. Residual plots are non-negotiable (K&S Ch. 20–25)

6 — Respect EPV rule

Include at most 1 variable per 10 events to avoid overfitting (K&S p. 246)

7 — Address missing data

Describe mechanism (MCAR/MAR/MNAR); use multiple imputation when appropriate (K&S p. 188)

8 — Report 95% CIs

P-values alone are insufficient — confidence intervals show magnitude and precision (K&S Ch. 8)

9 — Use STROBE checklist

Structured transparent reporting ensures reproducibility and peer review acceptance

10 — State limitations

Residual confounding, selection bias, measurement error, and generalisability (K&S Ch. 19)

10 Principles of Good Modelling Practice

Let theory guide variable selection

Use DAGs and prior knowledge — not stepwise p-values

Match the model to the outcome

Distribution of Y determines the model family

Always report crude + adjusted

Show the confounding adjustment transparently

Check every assumption

Violated assumptions → biased, misleading results

Respect the EPV rule

≥ 10 events per variable to avoid overfitting

Handle missing data explicitly

Describe mechanism and method — don't just delete

Validate your model

Bootstrap or external validation before publication

Report effect sizes + 95% CI

P-values alone are not enough — magnitude matters

Follow STROBE or TRIPOD

Structured reporting improves reproducibility

Acknowledge limitations

Residual confounding, generalisability, selection bias